

Rethinking Early Myeloma Care

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The Plasma Cell Disorders Continuum

Risk of progression to MM
about 10 in 100 per year
(first five years)

Treatment is indicated

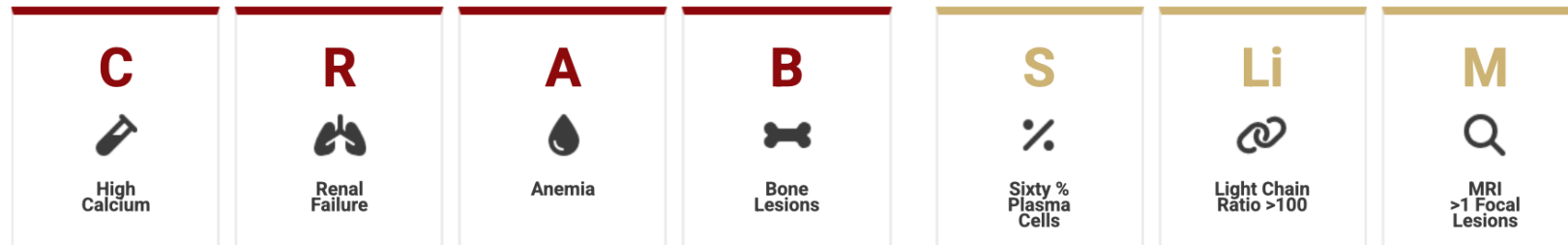
Smoldering multiple myeloma (SMM)

10–60% marrow plasma cells and/or an M-protein of 3 g/dL or more.
No organ damage, no ultra-high-risk markers.

Active multiple myeloma (MM)

Organ damage — bone, kidney, anemia, calcium —
and/or an ultra-high-risk markers.

The line between these two = CRAB-SLiM



The takeaway: how much disease is there, and is it doing harm?

Untangling My Thought Process

Smoldering myeloma

Should we treat early before damage even starts, or should we monitor closely?

Newly diagnosed MM transplant-eligible

Will transplant add any benefit?
How long should we treat for?

Newly diagnosed MM transplant-ineligible

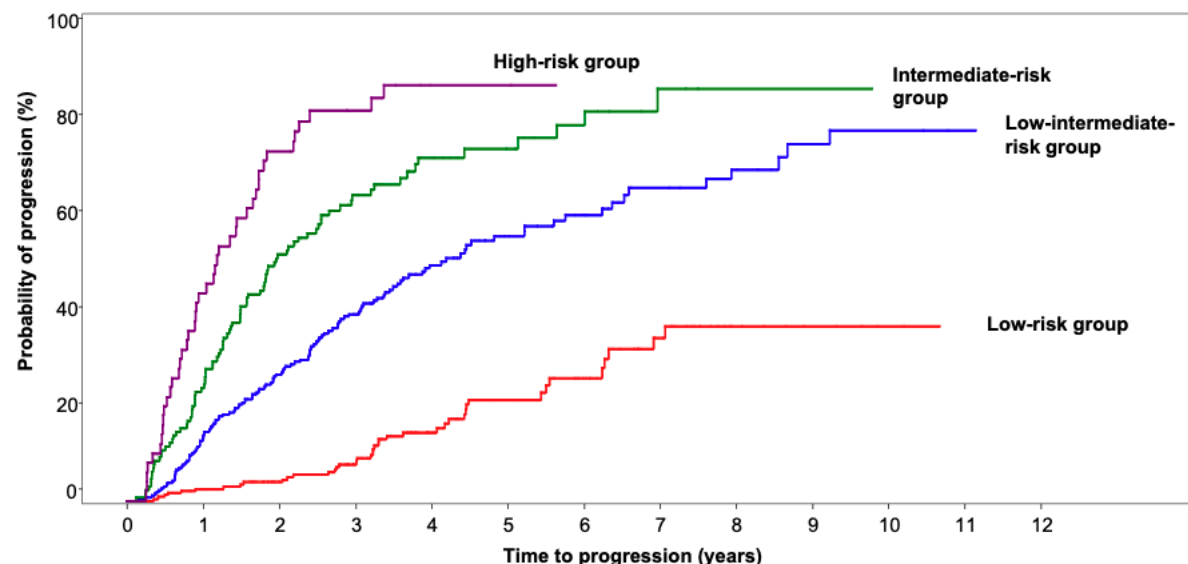
How much therapy can the patient tolerate?
How long should we treat for?

1. How fit (healthy) is the patient? Any social barriers present?
2. How much disease is there? What is the disease biology?
3. How is the disease behaving over time? How deep did the response go?

The takeaway: three different situations, one shared way of thinking. Whichever of the three buckets, these are the questions that drive our thought process and recommendations

Risk of Progression From SMM to Active MM

Risk factor	Score
FLC IU	
0–10 (reference)	0
>10–25	2
>25–40	3
>40	5
M-protein	
0–1.5 (reference)	0
>1.5–3	3
>3	4
BMPC	
0–15 (reference)	0
>15–20	2
>20–30	3
>30–40	5
>40	6
FISH abnormality	2



Risk Stratification groups	Total risk score	Hazard Ratio (95% CI)	Risk of progression (2 years)	# of patients
Low	0-4	Reference	3.8%	241 (35.0%)
Low-intermediate	5-8	7.56 (3.77 – 15.2)	26.2%	264 (38.3%)
Intermediate	9-12	17.3 (8.63 – 34.8)	51.1%	133 (19.3%)
High	>12	31.9 (15.4 – 66.3)	72.5%	51 (7.4%)

	Risk score
0	3.2
2	6.2
3	8.5
4	11.6
5	15.7
6	20.8
7	27
8	34.3
9	42.5
10	51
11	59.5
12	67.5
13	74.6
14	80.5
15	85.4
16+	89.2

The takeaway: know your numbers, know your risk of progression, and decide on your plan (shared decision making)

Should We Treat SMM Now, or Watch Closely?

Option A: Watch Closely (Low and Intermediate Risk)

Blood tests to monitor your calcium level, kidney function (creatinine), red cells (hemoglobin).

Monitor the M-protein and Free light chain level.

Clinic visits to discuss symptoms that could prompt repeat imaging test or a bone marrow biopsy.

Option B: Treat Now (High Risk)

We usually use this options for the patients who we think have the highest risk of progressing, $\geq 50\%$ in 2 years.

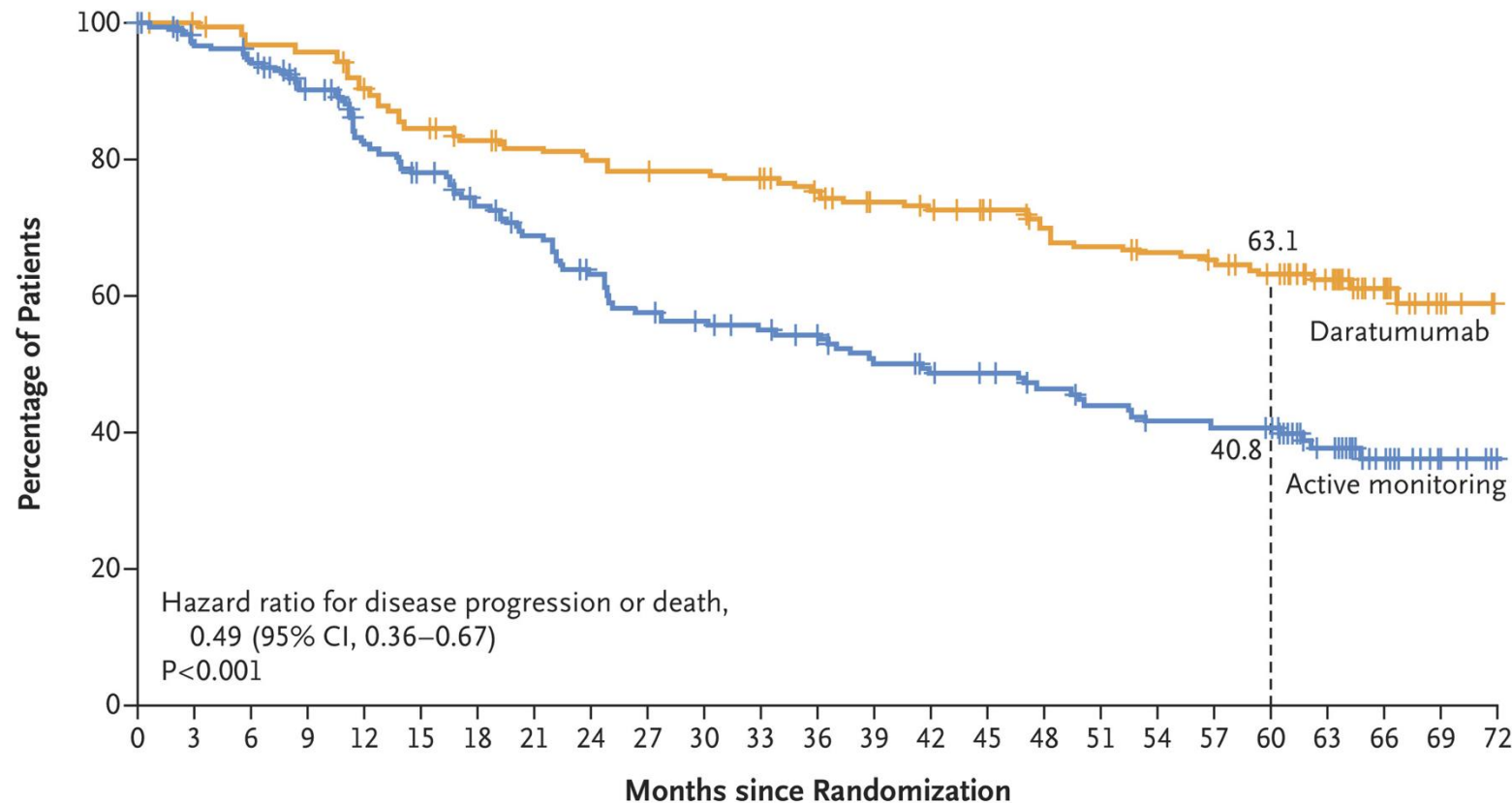
Treating early can delay - and in some cases prevent - the damage myeloma does. That is now proven.

Treating early also means giving medication for years to people who, maybe, were never going to be harmed or progress to myeloma.

This is a decision with two reasonable answers, and it belongs to you.

AQUILA: Treating Early Delays Progression

Progression-free Survival

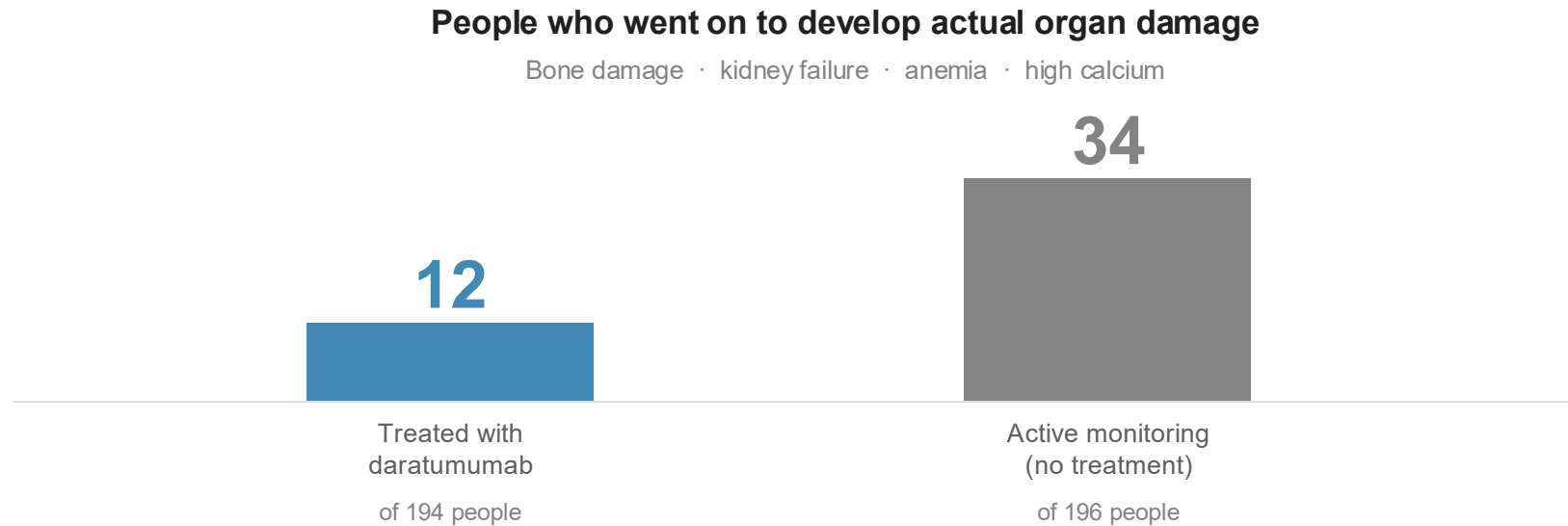


What you're looking at: the trial's main result. Time since joining runs along the bottom; each line is a group. The orange line is daratumumab, the blue line is monitoring alone.

Why it matters: the orange line stays higher throughout — after five years about 63 in 100 treated people were still free of progression, against about 41 in 100 who were monitored. The risk was cut roughly in half.

The takeaway: Daratumumab for a total of 3 years led to a real, durable difference. It is also a delay: the treatment is given for three years and then stopped, and 4 in 10 people who took nothing were still fine at five years.

What Matters Most – Preventing Organ Damage

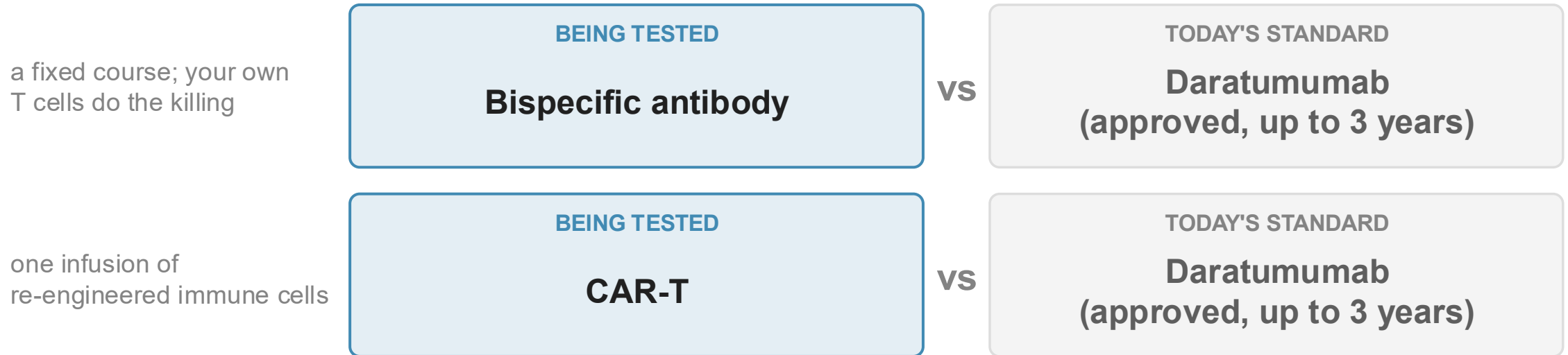


What you're looking at: the same trial, counted differently — “did something happen that a person would feel and care about”.

Why it matters: twelve treated people versus thirty-four monitored people went on to progress with bones lesions, kidney damage, anemia or high calcium.

And the cost: serious side effects in about 40 in 100 treated people versus 30 in 100 monitored; pneumonia in 3.6 versus 0.5 in 100; about 7 in 100 stopped early.

What's the Future in SMM?



The question is no longer “treat or watch.” It is whether something stronger beats daratumumab.

What changed: the comparison group moved. Before, trials tested treatment against watching. Now that daratumumab is approved, it has become the thing any new treatment has to beat.

The takeaway: a bispecific antibody works for a fixed course; CAR-T is given once. Both are being tested against daratumumab in high-risk smoldering disease — and a trial is the only way to get either one here.

What We Honestly Do Not Know

We do not know whether treating early helps people live longer. In AQUILA the survival difference rests on fifteen deaths versus twenty-six, and only about a third of the monitoring group who progressed went on to receive treatment for it.

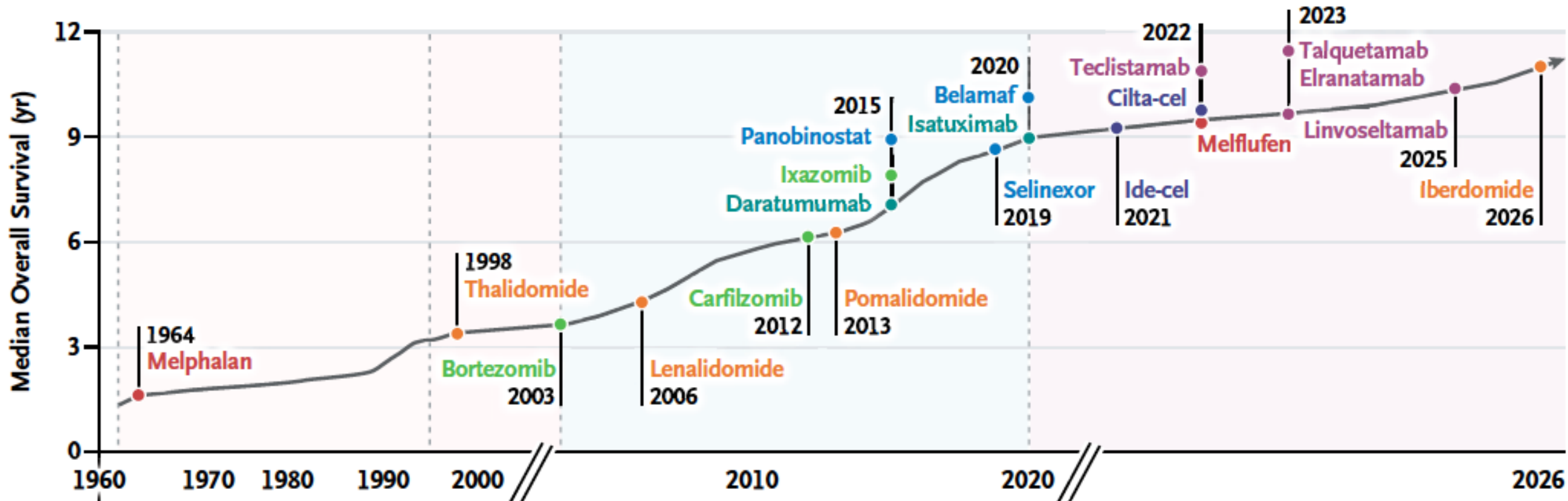
We cannot tell which individual will progress. Every model sorts populations, not people - and about one in five people enrolled in AQUILA would be classed as low-risk by today's criteria.

We are not curing most people yet. Some precursor trials now report undetectable disease in every patient - but follow-up is short, myeloma has returned from undetectable before, and the bar for treating people who might never have progressed is higher than in diagnosed disease.

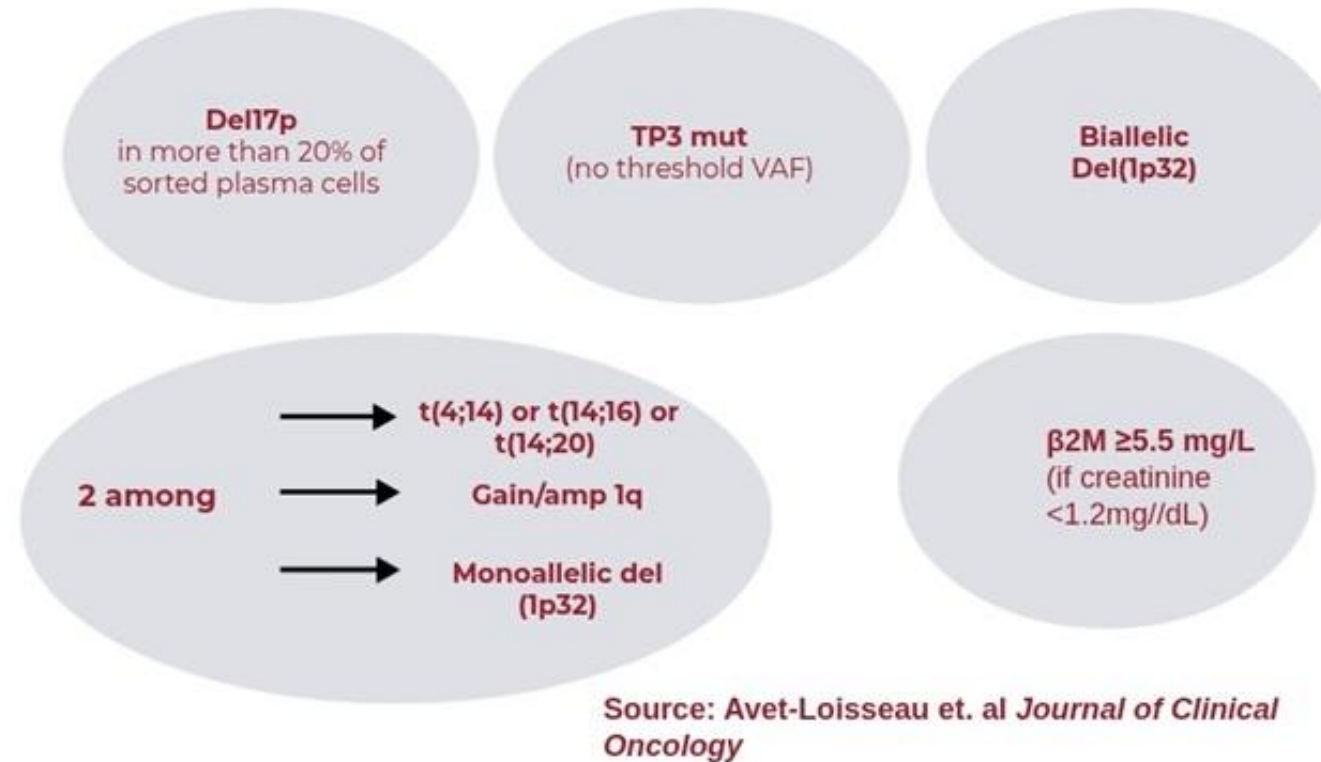
We do not know the long-term cost - infection risk, second cancers with the older drugs, and years of being a person in treatment rather than a person under observation.

We do not know whether treating early makes later treatment harder. That is why the newer trials measure what happens through the next line of therapy, not just the first.

What is The Treatment Landscape of MM?



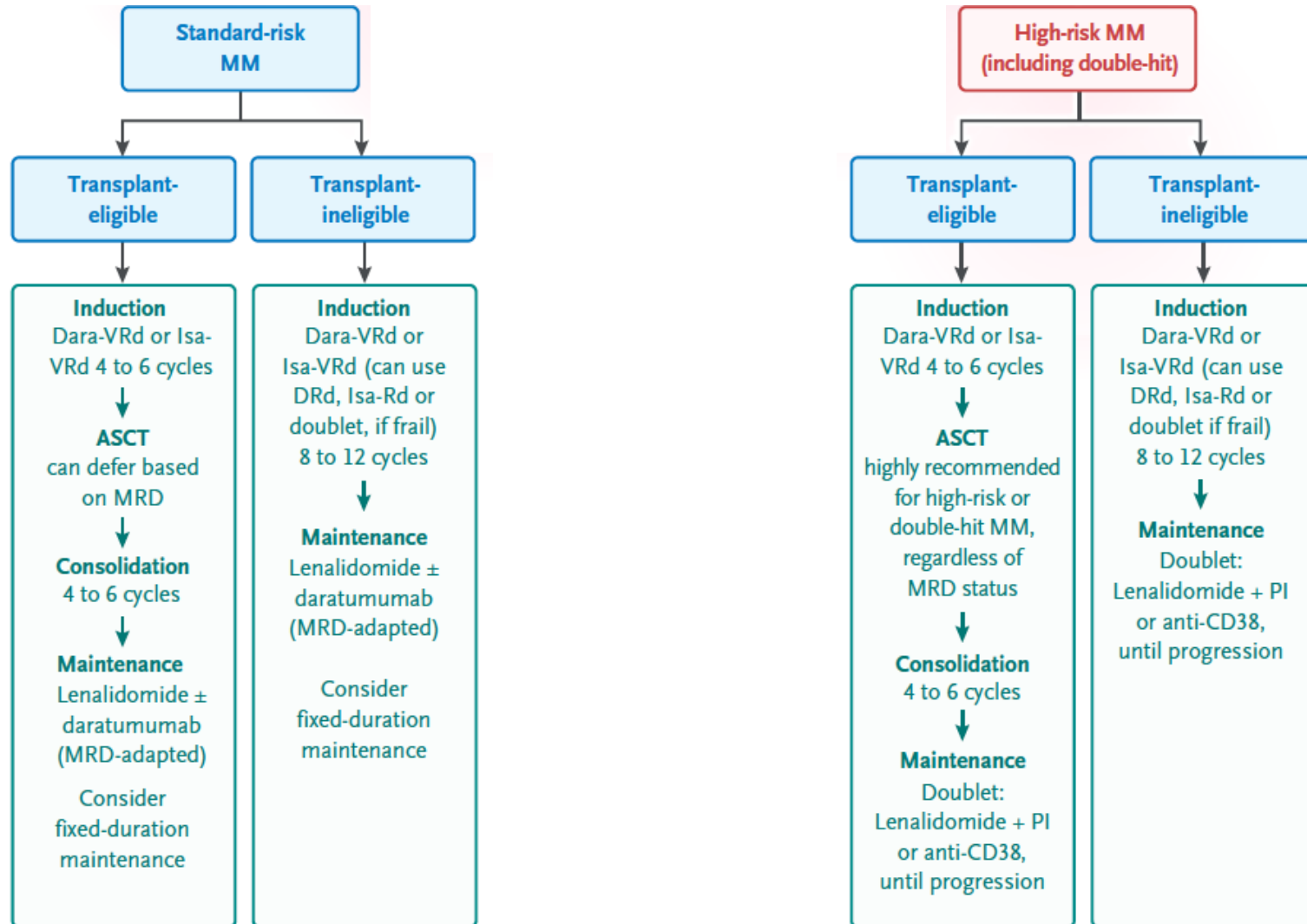
From ISS to RISS to IMS/IMWG Consensus Genomic Staging of High-Risk MM



Why it matters: Standard Risk vs High Risk is useful to plan treatment, it is not a prognosis or life-expectancy

The takeaway: if you have been told you are high risk, ask which system was used. The answer often changes the label.

Standard of Care for Newly Diagnosed MM



Could a Patient Tolerate a Bone Marrow Transplant?

- Functional status
- Psychological status
- Cognition
- Socioeconomic factors
- Comorbidities
- Nutritional status
- Sensory loss
- Polypharmacy



IMWG Frailty score

FIT

Age ≤ 75 years, ADL > 4 , IADL > 5 , and CCI ≤ 1

ASCT eligibility:

cardiac function (LVEF $> 40\%$)
liver function (bilirubin < 1.5 ULN, AST/ALT < 2.5 ULN)
pulmonary function (DLCO/FEV1 $> 40\text{-}80\%$)

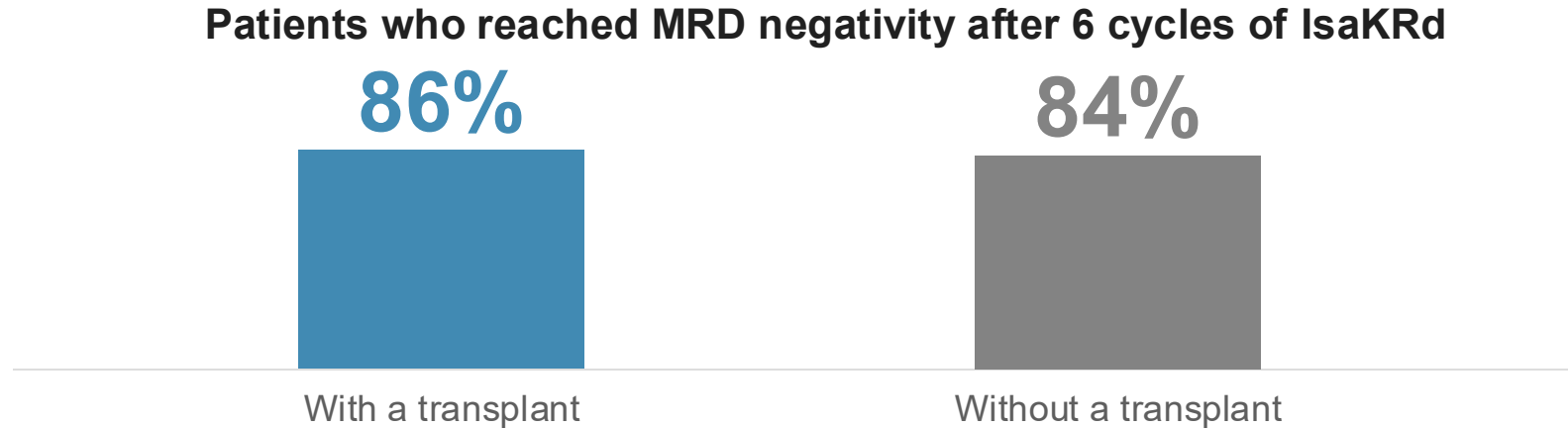
INTERMEDIATE-FIT

Age 76-80 years
or ADL ≤ 4
or IADL ≤ 5
or CCI ≥ 2

FRAIL

Age > 80 years regardless of ADL, IADL, CCI
or Age 76-80 years and either ADL ≤ 4 , IADL ≤ 5 , CCI ≥ 2
or Age ≤ 75 years and at least two of the following:
ADL ≤ 4 , IADL ≤ 5 , CCI ≥ 2

In Standard Risk MM, Transplant Is Now a “Choice”



Follow-up so far is only about sixteen months. Whether the two groups stay equal is not yet known.

What you're looking at: among people already in a very deep remission after four-drug treatment, the share who reached the deepest measurable level with a transplant and without one.

Why it matters: 86 versus 84 in 100 — no meaningful difference, and a second transplant added nothing. This is the trial that made deferring a defensible choice rather than a gamble.

The takeaway: a strong response with standard-risk chromosomes makes deferring reasonable; a weaker response, or high-risk or double-hit disease, still points to transplant.

Can We Stop Maintenance Therapy?

Standard risk, no upfront transplant: similar survival

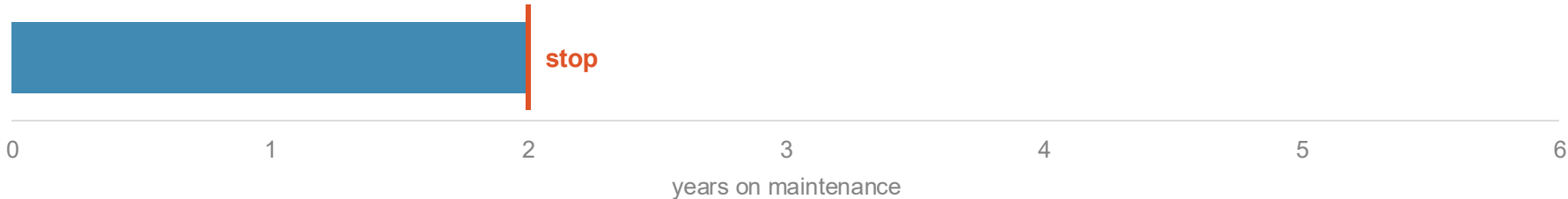
Continuous

lenalidomide until the disease progresses



Fixed duration

two years of lenalidomide, then stop



High-risk disease is the exception — there, maintenance continues until progression.

What you're looking at: two ways to run maintenance after treatment: keep going indefinitely, or stop at a planned two years.

Why it matters: for standard-risk patients who did not have an upfront transplant, a randomized trial found the fixed two years was enough — “forever” is no longer the automatic answer.

The takeaway: this is a different question from the last slide. That one asked how DEEP the remission is; this one asks how LONG you have been treated. High-risk disease is the exception either way.

Can We Replace Transplant Altogether?

CARTITUDE-6

NCT05257083
after Dara-VRd induction

BEING TESTED

CAR-T (cilta-cel)

VS

TODAY'S STANDARD

Stem-cell transplant

EILen (IFM 2025-01)

NCT06918002
after Dara-VRd induction

BEING TESTED

Bispecific antibody
(elranatamab + lenalidomide)

VS

TODAY'S STANDARD

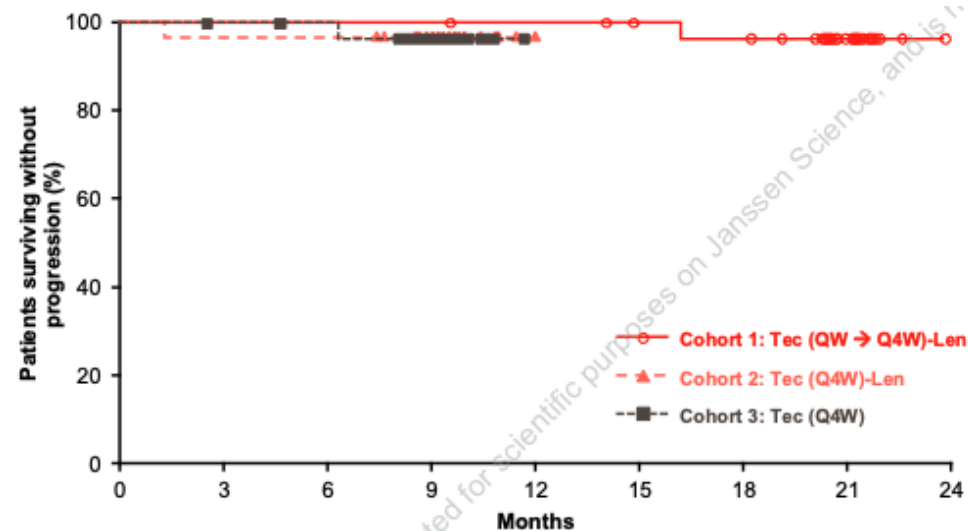
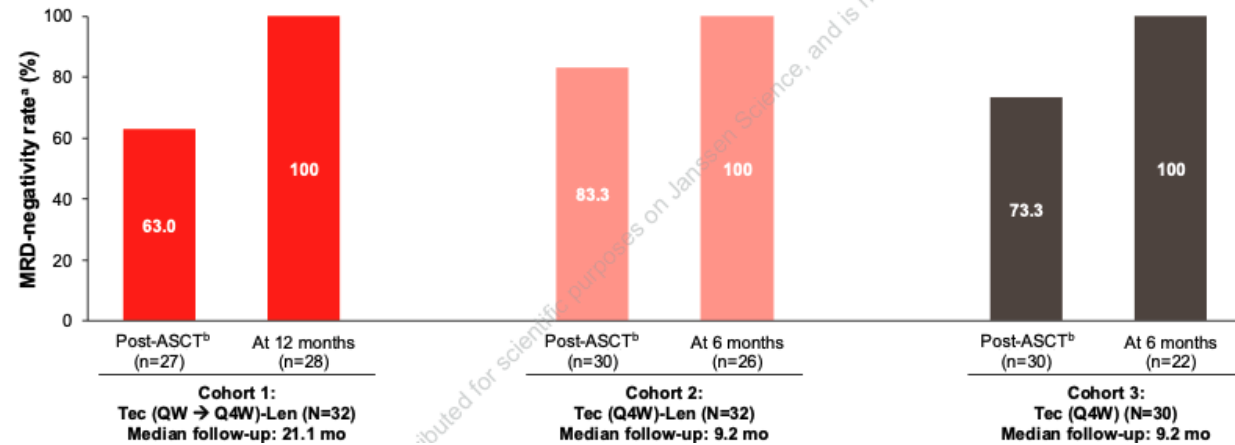
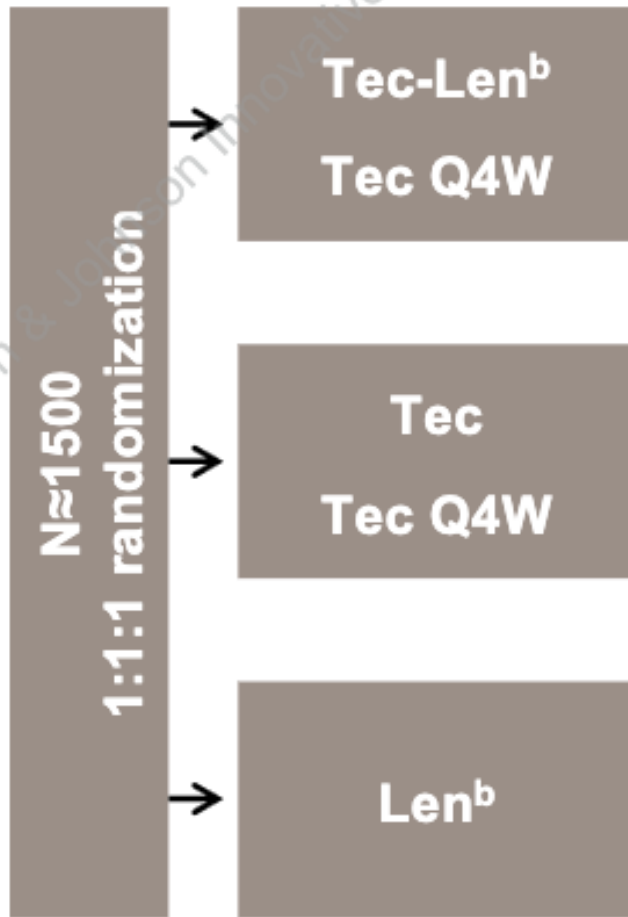
Stem-cell transplant

These are the trials that could retire the transplant for some people. Neither has reported yet.

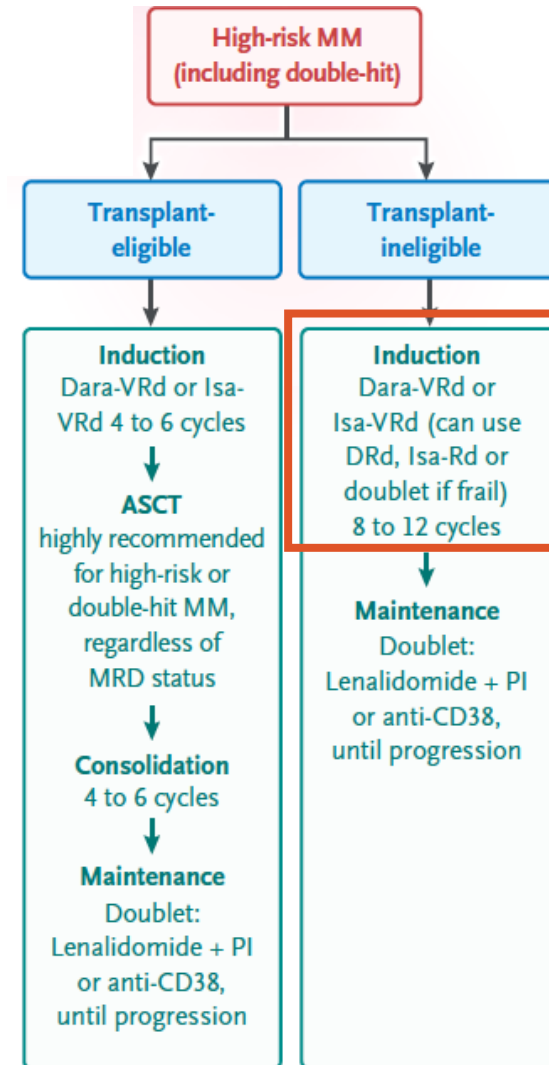
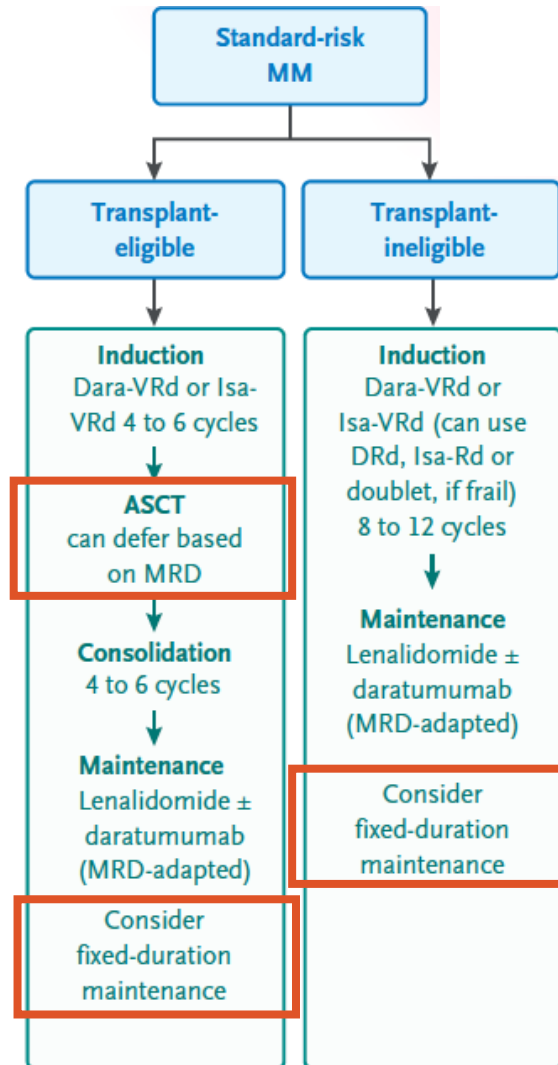
What you're looking at: two large trials in which people who could have a transplant are randomly assigned to get one, or to get an immune therapy instead — after the same four-drug start.

The takeaway: these are the trials that could retire the transplant for some people. Neither has reported yet, and both are worth asking your team about.

Can We Do Better Than Lenalidomide Maintenance?



Standard of Care for Newly Diagnosed MM



Personalizing Treatment Decisions in NDMM

Risk stratification - IMS/IMWG 2025 Consensus Genomic Staging criteria (functional high-risk, extramedullary disease and plasma cell leukemia)

Quadruplet induction - standard therapy for both transplant-eligible and fit, transplant-ineligible patients.

Autologous stem cell transplantation - frontline in high-risk MM, but patients with standard risk MM in MRD-negative responses may defer the procedure.

Maintenance therapy - doublet therapy for patients with high-risk disease; finite duration for patients with standard risk MM in sustained MRD negativity.

Questions?

Website: drmonge.com

 Anschutz | Cancer Center

